

Mediator Subunit MED16 Collaborates with UBP1-TFCP2 to Control Transcriptional Activation or Repression via Promoter Positional Specificity

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eLife Assessment

The characterization of a dissociable Mediator subunit implicated in cellular pathways, particularly lung alveolar function and HIV latency, would be conceptually interesting. The authors have preliminary evidence for a stable Med16 subcomplex that may regulate specific genes. This work is **useful** in that it points to interactions between Med16 and UBP1, but the evidence is preliminary and **incomplete**.

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Abstract

The Mediator complex is an evolutionarily conserved transcriptional coactivator with well-characterized structure and function, though the roles of its dissociable subunits remain incompletely defined. Here, we demonstrate that Mediator subunit MED16 dissociates from the core complex to form a subcomplex with transcription factors UBP1 and TFCP2, and this interaction modulates transcriptional activation or repression in a context-dependent manner. Using protein purification coupled with mass spectrometry, we identified MED16 as a binding partner of UBP1-TFCP2. Gene expression analyses showed that UBP1 interacts with MED16 to activate a subset of silenced genes involved in lung homeostasis, angiogenesis, and cell proliferation. Conversely, the MED16-UBP1 interaction suppresses HIV-1 transcription, thereby reinforcing viral latency. Mechanistically, MED16 and UBP1 cooperatively bind the HIV-1 transcriptional start site (TSS) to inhibit preinitiation complex assembly. Genomic-scale analyses further demonstrated that transcription is activated when the UBP1-TFCP2 binding motif is proximal to the TSS, but repressed when the motif overlaps the TSS. Collectively, our

findings identify a novel MED16-UBP1 interaction, define its dual role in transcriptional regulation, and highlight the therapeutic potential of targeting this axis in HIV-1 infection.

Introduction

The canonical model for eukaryotic gene transcription by RNA polymerase II (Pol II) is that transcription factors recruit co-activators and RNA Pol II. Together, they are assembled into the pre-initiation complex (PIC) along with general transcription factors (GTFs) at a gene promoter (1–8). Among several kinds of co-activators reported, the multi-subunit Mediator complex has emerged as an essential co-activator that promotes PIC formation and activity.

The Mediator complex is composed of ~30 subunits in mammals and structurally divided into four modules: Head, Middle, Tail, and Kinase. The scaffold subunit MED14 joins the Tail, Middle, and Head modules together to form the “core Mediator”. The tail module is composed of MED15, MED16, MED23, MED24, MED25, MED27, MED28, MED29, and MED30; and the main function of the tail module is to provide an interactional platform for transcription factors to recruit Mediator complex to gene promoters and enhancers for further regulation of gene expression (8–10). Among all tail subunits, MED16, MED23, MED24, and MED25 form a submodule that accounts for ~70% molecular weight of the tail module and creates a large platform for interactions of transcription factors (11). Previous studies have found MED23 interacts with multiple transcriptional factors and controls diverse gene expression programs at multiple stages of transcription processes (12–15). It prompts us to question whether other subunits in this submodule also play similar roles in transcriptional regulation, especially MED16 that is structurally conserved from yeast to mammals and acts as a critical regulator in yeast and plants. Sin4 (yeast MED16) plays both positive and negative roles in the transcriptional regulation of many genes in yeast, and loss of Sin4 alters chromatin accessibility globally (16). In plant cells, MED16 controls endoreduplication and cell growth by repressing the expression of APC/C activator genes *CCS52A1/A2* (17–21). MED16 also controls plant cold response, chemical and wound response, and iron homeostasis at transcriptional levels (17–21). However, the functions of MED16 in mammalian cells are poorly understood. Among a few reports, MED16 was reported as a key factor in NRF2-activated gene expression via its direct interaction with NRF2. Knockout of MED16 reduces NRF2-dependent anti-oxidative gene expression (21, 22). Additionally, MED16 was down-regulated in papillary thyroid cancer, leading to increased TGF- β signaling and radioiodine resistance (23). Though the exact molecular mechanisms by which MED16 regulates gene expression are poorly understood. Its roles in animal development and diseases are barely investigated.

UBP1 (as known as LBP-1) and TFCP2 (as known as LSF) belong to a subfamily of the TFCP2/Grainyhead transcription factors family. The amino acid sequences of TFCP2 and UB1P share 88% identity, while TFCP2 and TFCP2L1 share 70% identity (24–26). UB1P and TFCP2 bind to the same consensus sequence CNRG-N6-CNR(G/C) (27). They function as a transcription activator or repressor to regulate a number of viral and cellular genes, including mouse *α -Globin*, *MMP-9*, *CYP11A1*, *IL-4*, *SV-40*, and *HIV-1* (28–32). UB1P and TFCP2 are involved in various biological processes, including embryonic development, angiogenesis, tumorigenesis, and HIV-1 latency (28, 29, 31–38). There is a canonical UB1P binding motif within the HIV-1 core promoter which overlaps with the transcription start site (TSS). Previous studies have reported that UB1P inhibits the binding of TFIID to the TATA-box (32). On the other hand, UB1P restricts HIV-1 transcription at the level of elongation (33). UB1P also has been demonstrated to inhibit HIV-1 transcription by recruiting YY1 and HDAC1 (34). However, how UB1P and TFCP2 regulate cellular gene expression, and what are the genome-wide target genes of UB1P and TFCP2 still remain unclear.

In this study we identified a tripartite complex consisting of UBP1, TFCP2, and MED16 utilizing gel-filtration chromatography and immunoprecipitation-mass spectrometry (IP-MS). MED16 interacts with UBP1 to activate expression of an array of endogenous genes, including several pulmonary surface molecules. In addition, MED16 cooperates with UBP1 to inhibit HIV-1 transcription at the HIV-1 transcription initiation site. At the genomic scale, we found out that UBP1 activates gene expression when the UBP1-TFCP2 binding motif is located near TSS, but inhibits gene expression by obstructing normal PIC formation when the UBP1-TFCP2 binding motif overlaps with TSS. Based on these results, we propose a dual functional model of MED16 in gene transcriptional control, and provided the underlying mechanisms of UBP1-repression on the HIV-1 genome, which implying a strategy for HIV-1 therapeutics.

Result

Identification of the dissociable MED16 from the Mediator complex

The Mediator complex consists of large intrinsically disordered regions (IDRs) and lacks predictable functional motifs (39, 40). The Mediator complex forms liquid-liquid phase separation condensates at super-enhancers. These IDR-mediated condensates can be disrupted by 1,6-hexanediol (1,6-HD) (41, 42), an aliphatic alcohol, via inhibition of weak hydrophobic protein-protein interactions (43–45).

To investigate whether 1,6-hexanediol affects the integrity of the Mediator complex, we utilized three antibodies against different Mediator subunits (CDK8, MED1 and MED12) to immunoprecipitate the Mediator complex from the 293T whole cell lysates, respectively. The immunoprecipitates were washed by washing buffers containing different concentrations (0%, 5%, 10%) of 1,6-hexanediol and blotted by antibodies against different Mediator components. Most of the detected subunits show modest decreases, whereas MED16 exhibits a pronounced reduction in response to increasing 1,6-hexanediol treatment (Fig 1A). To further confirm this observation, a similar experiment was performed in HeLa nuclear extract (NE) with 2,5-hexanediol as a control treatment. As an isomer of 1,6-hexanediol, 2,5-hexanediol does not interfere weak hydrophobic interaction due to the different positions of oxyhydrils. Once again, the amount of immunoprecipitated MED16 was easily decreased by 1,6-hexanediol treatment but not by 2,5-hexanediol treatment (Fig 1B), suggesting that MED16 is an easily dissociable subunit.

We then immunoprecipitated the Mediator complex with MED1 antibody in high salt condition to test whether the binding of MED16 to Mediator complex is charge dependent. The amount of immunoprecipitated MED16 remained unaltered under 150 mM, 300 mM and 500 mM NaCl conditions (Fig 1C). These observations suggest that MED16 is integrated into the Mediator complex through weak hydrophobic interaction but not ionic interaction.

To directly examine if MED16 is dissociable from the Mediator complex, we applied HeLa NE to Superose 6 gel-filtration column and separated Mediator subunits within each fraction that were then detected by immunoblotting. Most Mediator subunits, including MED16, were co-eluted at the fraction at about 2,000 kDa, suggesting these fractions contain the integrated Mediator complex (Fig 1D). However, a significant amount of MED16 was eluted at the fraction between 443 kDa and 150 kDa, away from other core Mediator subunits, indicating MED16 within this fraction is not part of the core Mediator complex (Fig 1D). MED16 subunit may be able to dissociate from the Mediator complex because the interaction between Mediator and MED16 is an unstable hydrophobic interaction.

MED16 interacts with UBP1 and TFCP2 in vivo and in vitro

Because MED16 is about 95 kDa and eluted at the HeLa NE gel-filtration fractions between 443 kDa and 150 kDa, MED16 may form a higher molecular mass complex with other proteins. To identify the proteins associated with MED16, we performed immunoprecipitation-coupled mass spectrometry (IP-MS) analysis using antibodies against four Mediator subunits, including MED1, MED12, CDK8 and MED16, respectively. Forty-two proteins were identified as exclusively interacting with MED16 in IP-MS data. Among these, two dimeric transcription factors, UBP1 and TFCP2 (**Fig S1** [↗](#)), were found to be co-eluted with the dissociable MED16 in gel filtration chromatography (**Fig 1D** [↗](#)). This suggests that UBP1 and TFCP2 interact with the dissociated MED16 (**Fig S1** [↗](#)).

To determine whether UBP1 and TFCP2 form a complex with MED16, an antibody specific to MED16 was used to capture MED16 and its associated proteins in HeLa NE or HeLa NE gel-filtration fractions between 443 kDa and 150 kD. MED1, MED23, TFCP2 and UBP1 were co-immunoprecipitated with MED16 in HeLa NE, but only TFCP2 and UBP1 were co-immunoprecipitated with MED16 in NE fractions between 443KDa and 150KD (**Fig 1E** [↗](#)). Reciprocally, anti-TFCP2 antibody is also able to pull down MED16 in either HeLa NE or 443KDa-150KDa fractions (**Fig 1F** [↗](#)). YY1 and HDAC1 are known to complex with TFCP2 and UBP1 to suppress HIV-1 transcription ([34](#) [↗](#), [46](#) [↗](#), [47](#) [↗](#)). YY1 was pulled down in both samples, whereas only a small amount of HDAC1 was pulled down from HeLa NE but not from the 443KDa to 150KD fractions (**Fig 1E** [↗](#)).

Because TFCP2 and UBP1 can be co-immunoprecipitated with MED16 and other core Mediator subunits, we examined whether TFCP2 and UBP1 interact with the Mediator complex. Both the TFCP2 and UBP1 were co-immunoprecipitated with Mediator complex under the 2,5-hexanediol treatment, whereas their association with Mediator complex was abolished under treatment with 1,6-hexanediol (**Fig 1B** [↗](#)). These results suggest that TFCP2 and UBP1 are complexed with MED16 in vivo and dissociate from the Mediator complex, possibly along with the dissociation of MED16.

To extend the analysis of the MED16 subcomplex, a domain-mapping experiment with deletions of multiple structural domains of MED16 was carried out. The N-terminal of MED16 contains nine WD40-repeat (WDR) domains that form a seven-bladed β -propeller structure (**Fig 2A** [↗](#)) ([11](#) [↗](#), [48](#) [↗](#)). This structure appears to be crucial for MED16's integration into the Mediator complex, as MED16 with WDR-deletion shows a dosage-dependent effect on its association with Mediator complex (as indicated by MED1 here); in details, this association gradually decreased as more WDR domains are truncated (**Fig 2B** [↗](#)). The C-terminal of MED16 harbors an $\alpha\beta$ -domain with 157 amino acids ([11](#) [↗](#), [48](#) [↗](#)) and the MED16 truncated mutant without $\alpha\beta$ -domain failed to interact with UBP1-TFCP2 (as indicated by TFCP2 here) (**Fig 2B** [↗](#)). Therefore, the N-terminal WDR domain of MED16 is crucial for its integration into the Mediator complex, while the C-terminal $\alpha\beta$ -domain is essential for interacting with UBP1-TFCP2.

We conducted a GST pull-down experiment to assess the direct interaction between individual transcription factors and MED16. We expressed and purified UBP1, TFCP2, YY1, and TFCP2L1 that were fused with 6×His tag using bacterial (*E.coli*) expression system (**Fig 2C** [↗](#)). Transcription factor CP2 like 1 (TFCP2L1), a paralog of TFCP2, was included for comparison. Due the solubility issue with full length MED16, we purified only GST-fused C-terminal $\alpha\beta$ -domain of MED16 (GST-M16 $\alpha\beta$) that comprises 157 amino acids. GST-M16 $\alpha\beta$ exhibited the strongest binding to recombinant UBP1 (**Fig 2C** [↗](#)), suggesting that UBP1 binds to MED16 directly.

The amino acid sequences of UBP1 and TFCP2 share 88% identity, we investigated why UBP1, but not TFCP2, directly binds to MED16. Through comparison of the amino acid sequences of UBP1 and TFCP2, we found that a 36-amino-acid-long sequence between the DNA binding domain and SAM motif is specific to UBP1 (UBP1-SP) (**Fig 2D** [↗](#)). To test if UBP1-SP is responsible for interacting with

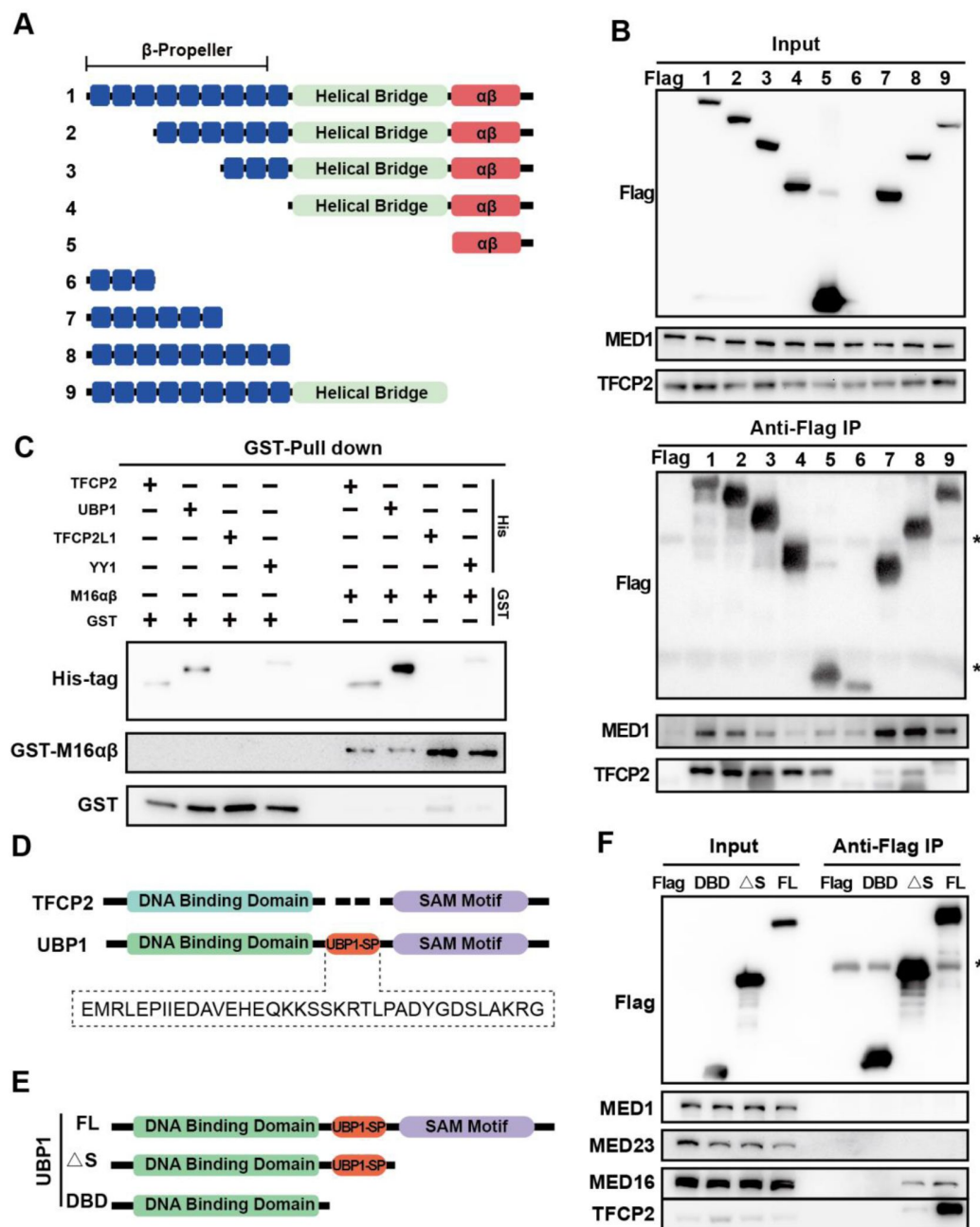


Figure 2.

MED16 αβ-domain interacts with UB1 specific 36 amino acids sequence in Vivo and in Vitro.

(A) Schematic of Flag-MED16 deletion mutants. The WDR domains are shown as blue boxes. "1" represent full-length MED16 and "2" to "9" represent different truncations of MED16. All proteins were labeled with Flag-tag and expressed in 293T cell line as panel (B) indicated. (B) Domain mapping experiment of MED16. Flag-MED16 deletion mutants were coimmunoprecipitated with endogenous MED1 and TFCEP2 using anti-Flag antibody. Asterisks indicate the IgG heavy and light chains. (C) Purified GST and GST-MED16 αβ-domain proteins were incubated with 6×His-tag TFCEP2, UB1, TFCEP2L1, and YY1 at 4°C. After washing the interaction protein were eluted by boiling and immunoblotted with antibodies against GST-tag and His-tag. (D) Amino acids sequence analysis of UB1 and TFCEP2 protein, a 36 amino acids UB1-specific sequence (UB1-SP) as shown below. (E) Schematic of Flag-UB1 deletion mutants. FL represents full-length UB1, ΔS represents SAM motif truncated UB1, and DBD represents UB1 DNA binding domain. (F) Domain mapping experiment of UB1. Flag-UB1 deletion mutants were coimmunoprecipitated with endogenous MED1, MED23, MED16, and TFCEP2. Asterisks indicate the IgG heavy and light chains.

MED16, a UBP1 domain-mapping experiment was performed (**Fig 2E** and **2F**). The full-length Flag-tagged UBP1 successfully pulled down endogenous TFCP2 and MED16. In contrast, the SAM motif-deleted UBP1 truncated mutant could only bind to MED16 modestly, and failed to bind to TFCP2. DNA binding domain of UBP1 (DBD) didn't bind either MED16 or TFCP2 at all. These results indicate that MED16 complexes with UBP1 and TFCP2 in vivo and in vitro, and that the UBP1-SP domain is crucial for the interaction with MED16.

MED16 is required for UBP1 activation of gene transcription

Because the interaction between transcription factors and Mediator subunits is critical for target gene activation, we analyzed if MED16 is required for UBP1-driven gene transcription using reporter gene assay (**Fig 3A**). In the UBP1 overexpression assay, the full-length UBP1 protein significantly promoted transcriptional activation. However, the DNA-binding domain of UBP1 (DBD, 1-274aa) alone was unable to activate the reporter. The truncated UBP1 mutant (Δ S, 1-369aa), which contains the MED16 binding motif but lacks the TFCP2-binding SAM motif, exhibited about half the transcriptional activation when compared to the full-length UBP1 (FL) (**Fig 3B**).

To further confirm that MED16 is required for the transcriptional activity of UBP1, a HeLa cell line with MED16 knockdown was generated using viral-mediated shRNA targeting MED16, along with a negative control cell line (sh-NC). The UBP1-responsive reporter showed similar baseline of transcriptional activity in the two cell lines. The reporter was activated about 4-fold by overexpressing UBP1 in the negative control cell line, whereas knockdown of MED16 attenuated the activation to approximately two-fold (**Fig 3C**).

The coding sequences of human *MED16* and Mouse *Med16* share about 90% identity. Thus, we further established a *Med16* knockout cell line generated using CRISPR-Cas9 gene editing method in a mouse pancreatic cancer cell line called HT. Five *Med16* knockout clones (ko16a, ko16b, ko16c, ko16d, & ko16e), with different frameshift mutations were selected and verified by sequencing and Western-Blotting (**Fig 3F**). All *Med16*-KO clones displayed 2-7-fold reduced reporter activity under ectopic UBP1 activation when compared to the wildtype HT cells, and *Med16* KO also exhibited decreased basal level of UBP1-reporter activity without UBP1 overexpression (**Fig 3E** and **3F**). These results confirmed the specific requirement of MED16 for UBP1 transcriptional activation.

To further verify the specific regulation of UBP1 activity by MED16, we performed a titration experiment by co-transfected low and high doses of MED16 or MED16 $\Delta\alpha\beta$ (**Fig 2A**, truncation "9") with the UBP1-luciferase reporter. It has been reported that overexpression of Mediator subunits can result in inhibition of transcription owing to competition between the overexpressed subunits and the Mediator complex for the transcriptional activator (15, 49-52). In agreement with these results, UBP1 luciferase reporter activity tended to be increasingly repressed by MED16 in a dosage-dependent manner. In contrast, MED16 $\Delta\alpha\beta$, which cannot interact with UBP1, failed to repress the luciferase reporter activity (**Fig. 3G**). These findings highlight a specific coupling between MED16 and UBP1 in activating gene transcription.

The UBP1-TFCP2 hetero-dimerization is important for UBP1-responsive reporter transcription as it is shown above (**Fig 3B**). We then tested whether UBP1, TFCP2, and MED16 had a combinatorial effect on gene transcription. UBP1 and TFCP2 co-overexpression had better (from 9.8 times of basic level to 12 times of basic level) transcriptional activity than UBP1 alone, but MED16 co-overexpression reduced UBP1's activity in all of our combinations (**Fig 3H**). Together, the results of the UBP1-responsive reporter suggested that MED16 is required for UBP1 activation of gene transcription.

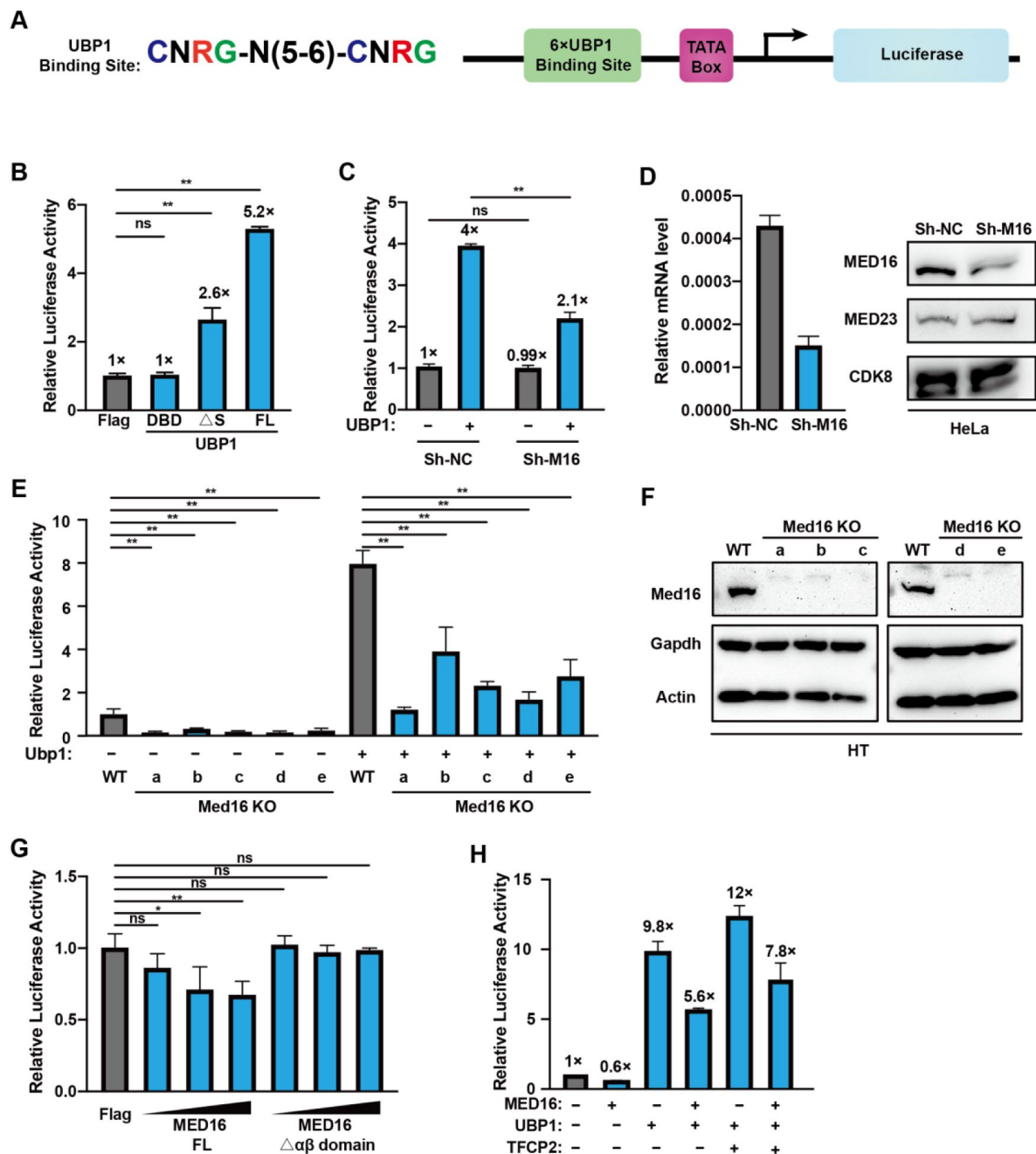


Figure 3.

MED16 is required for UBP1 activation of gene transcription

UBP1 reporter assay (A) with co-transfection of UBP1, MED16, and their mutants in various cell lines. In HeLa cells (B), 300 ng of UBP1 DNA binding domain (DBD), SAM motif-deleted mutant (Δ S), or full-length UBP1 (FL) plasmid was transfected with the reporter. (C) In MED16 knockdown and control HeLa lines generated by shRNA, 300 ng of UBP1 plasmid or control vector was co-transfected with the UBP1 reporter. Knockdown efficiency was confirmed by qPCR and Western blotting (D). (E) In wild-type and MED16 knockout mouse pancreatic cancer HT cells, the UBP1 reporter was co-transfected with 300 ng of mouse UBP1 plasmid or vehicle control. Knockout efficiency was validated by Western blotting (F). (G) Effect of increasing MED16 doses (0.3 μ g, 0.5 μ g, 1 μ g) or α β -domain truncated MED16 on the UBP1 reporter. (H) Effects of MED16, UBP1, and TFPC2 combinations on UBP1 reporter activity. Firefly luciferase activity was normalized to Renilla luciferase activity. The luciferase expression levels between groups were compared using two-tail unpaired Student's t-test. The data are presented as the means $\pm$ SDs from $n = 3$ independent experiments. *: $p < 0.05$, **: p -value < 0.01 and "ns": no significant.

Analysis of the UBP1 and MED16 co-regulated transcriptome

We went on to profile the UBP1 and MED16 co-regulated genes on a genome-wide scale using RNA-seq via comparison of the transcriptomes between MED16-knockdown and control HeLa cells with or without ectopic UBP1 expression. A total of 217 genes were activated (>1.5-fold), and 60 genes were repressed (>1.5-fold) in UBP1 overexpressing HeLa cells (**Fig 4A**). Among the 217 UBP1-activated genes, 198 were repressed by shMED16 knockdown (**Fig 4B** and **4C**). This result demonstrated that UBP1 and MED16 collaborate to regulate transcription in a genome-wide scale, further supporting that MED16 is required for UBP1 transcription activities.

To explore the potential biological processes that are co-regulated by MED16 and UBP1, 198 co-regulated genes by MED16 and UBP1 were subjected to gene ontology (GO) analysis (**Fig 4D** and **4E**). Interestingly, GO analysis revealed that the genes involved in the respiratory gaseous exchange process were regulated by MED16 and UBP1, such as *SFTPA1*, *SFTPA2*, *SFTPB*, and *SFTPC* (**Fig 4E**). These genes encode pulmonary surfactant associated proteins that promote alveolar stability by lowering the surface tension at the air-liquid interface in the peripheral air spaces. They are normally only expressed in alveolar epithelial cells and silenced in HeLa cells. However, highly matched UBP1 motifs were identified in their promoters (**Fig 4F**), leading to their activation when UBP1 was overexpressed. In our analysis, angiogenesis and cellular response to VEGF stimulus are also enriched in GO analysis (**Fig 4D**), which is consistent with the previous report that UBP1 plays a critical role in the regulation of extraembryonic angiogenesis (29, 53). Collectively, these results demonstrated that UBP1 and MED16 may collaborate to activate a target array of endogenous genes expression.

MED16 collaborates with UBP1 to inhibit HIV-1 transcription

Beyond their roles in transcriptional activation, UBP1 and TFDP2 have been reported to inhibit HIV-1 transcription through complex mechanisms (32–34). We questioned whether the MED16 also play a role in HIV-1 transcription repression. Thus, we constructed an HIV-1 reporter that consists of the HIV-1 core promoter (–78 to +60) and luciferase CDS (**Fig 5A**).

Previous research has shown that overexpression of UBP1 or YY1 inhibits HIV-1 transcription (34, 47). Consistently, we found our HIV-1 reporter's activity was reduced by overexpressing either UBP1 or YY1 in HeLa cells, validating our reporter's response to UBP1 and YY1 (**Fig 5B**). We then tested if MED16 participates the UBP1-driving HIV-1 transcription repression. We observed that overexpression of MED16 inhibited HIV-luciferase activity in a dose-dependent manner (**Fig 5C**). However, the MED16 mutant with the UBP1 interacting $\alpha\beta$ -domain deleted failed to inhibit HIV-1 reporter despite similar increase in the dosage of transfected MED16 $\Delta\alpha\beta$ (**Fig 5D**). This result demonstrated that MED16-mediated HIV-1 inhibition depends on its binding to UBP1.

To see if this HIV-1 inhibition is MED16 specific, MED23 and MED24 that are part of the Mediator complex tail module neighboring MED16 were cotransfected with the HIV-1 reporter, respectively. Neither MED23 nor MED24 showed any effect on the activity of the HIV-1 reporter (**Fig 5E**), thus underscoring the specific role of MED16 in repressing HIV-1 transcription. Moreover, MED16 knockdown by sh-RNA also modestly increased HIV-1 luciferase activity (**Fig 5F**). Collectively, these results demonstrated that MED16 collaborates with UBP1 to inhibit HIV-1 transcription.

UBP1 binding site determines the UBP1-MED16 inhibition in HIV-1 transcription

In a previous study, UBP1 was shown to inhibit HIV-1 transcription by recruiting YY1 and HDAC1 (Coull et al., 2000). To investigate if MED16 employs a similar mechanism to repress HIV-1 transcription, we transfected HeLa cells with an HIV-1 reporter and treated them with

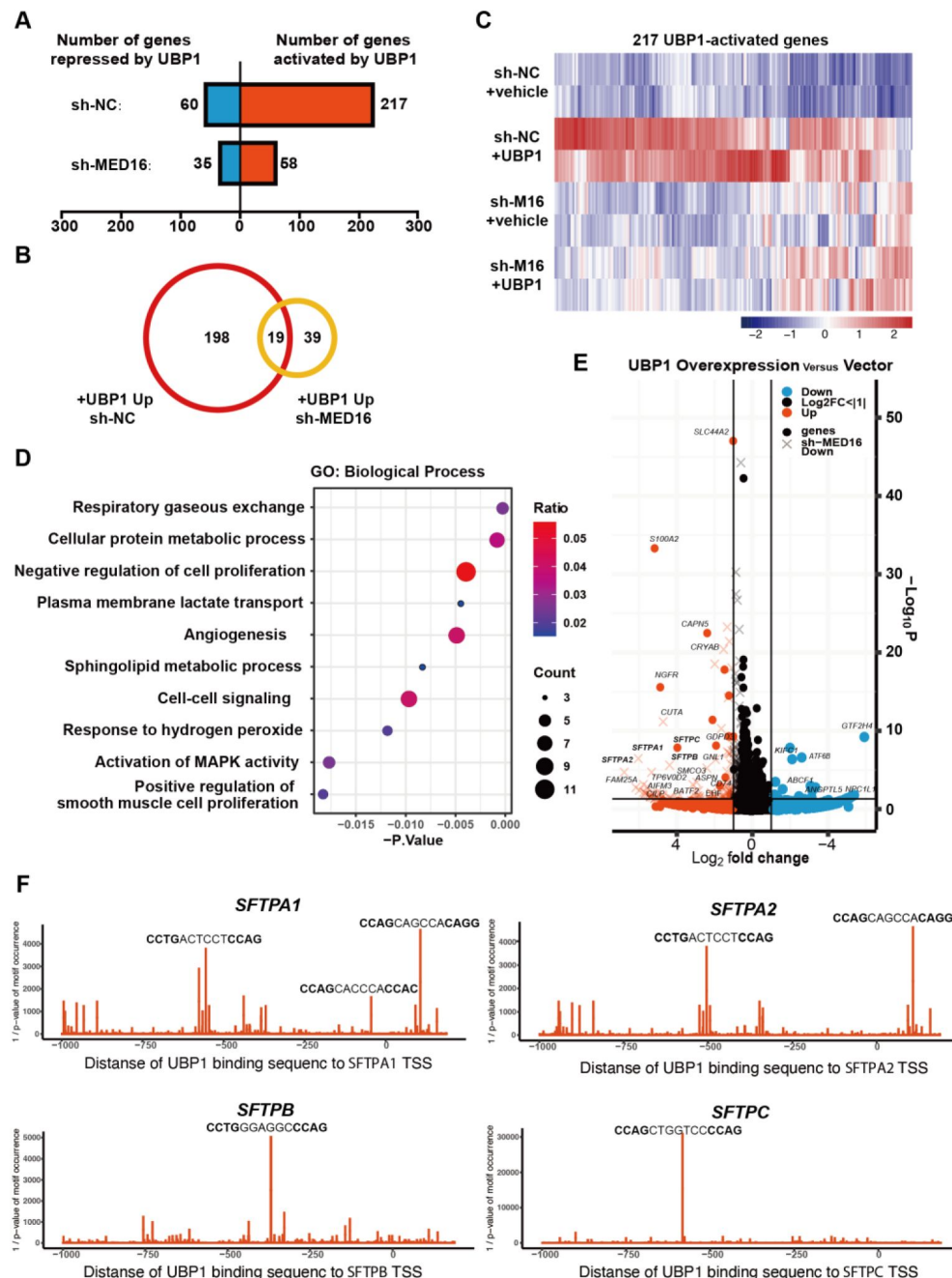


Figure 4.

UB1 and MED16 co-regulated transcriptome analysis

(A) The number of genes repressed or activated by UB1 overexpression in the sh-NC and shMED16 HeLa cell line ($FC > |1.5|$, $p \leq 0.05$). (B) Venn diagram of UB1-activated genes in the sh-NC HeLa cell line and the shMED16 HeLa cell line ($FC > |1.5|$, $p \leq 0.05$). 198 genes which were activated by UB1 overexpression but abolished by sh-MED16 were defined as UB1 and MED16 co-regulated genes. (C) Heatmap showing the RNA-seq analysis of 217 UB1-activated genes in sh-MED16 and sh-NC cell line treated with or without UB1 overexpression. UB1-activated genes were defined in sh-NC cell line under UB1 overexpression at fold change $> |1.5|$, p -values ≤ 0.05 . (D) GO analysis of 198 UB1 and MED16 co-regulated genes. The top 10 biological process items were presented. (E) Volcano map analysis of differentially expressed genes in sh-NC HeLa cell line with or without UB1 overexpression. Genes failed to be activated by UB1 in the sh-MED16 HeLa cell line we're indicated by "x". (F) UB1 motif occurrence analysis of *SFTPA1*, *SFTPA2*, *SFTPB* and *SFTPC*. Motif occurrence were showed within the promoter region of the genes (-1000 bp to 200 bp relative to the TSS).

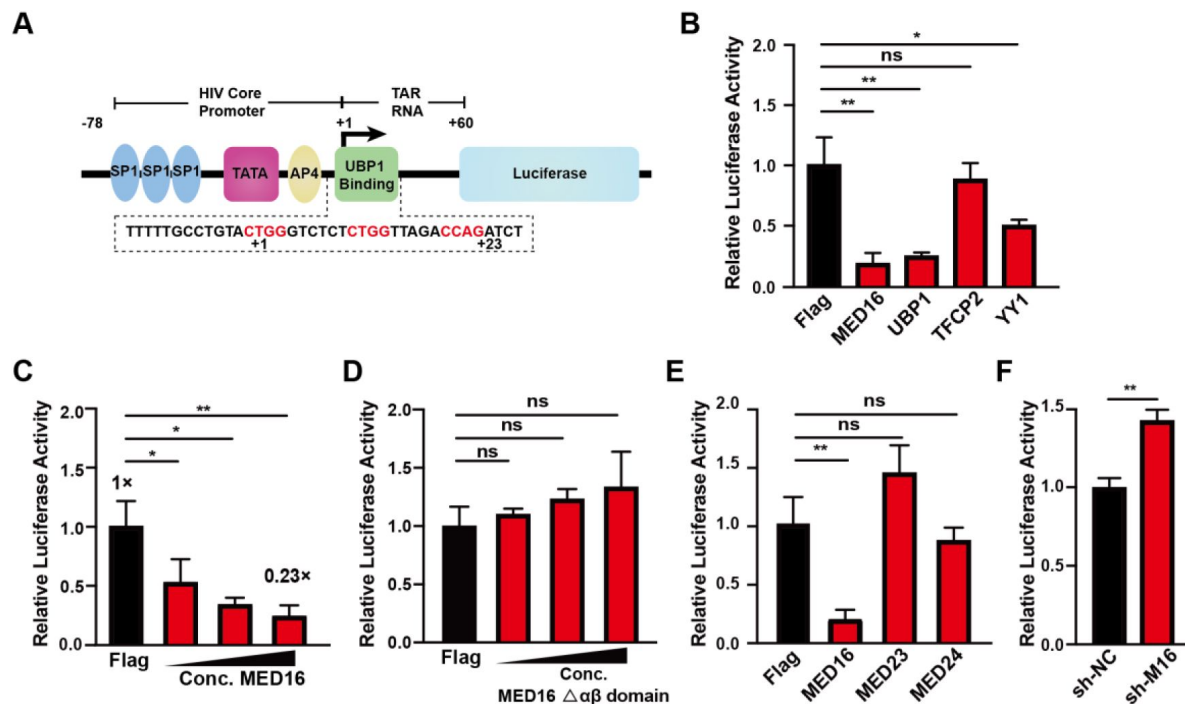


Figure 5.

MED16 cooperates with UBP1 to inhibit HIV-1 reporter expression

(A) Schematic illustration of the HIV-1 reporter, with the HIV-1 core promoter and +1 to +60 TAR RNA sequence cloned upstream of the luciferase coding sequence (CDS). (B) The HIV-1 reporter was co-transfected with 1.5 μg of MED16, UBP1, TFCP2, YY1, or Flag-tag control plasmids in 293T cells. (C and D) Increasing amounts (0.7 μg, 1 μg, 1.5 μg) of MED16 or αβ-domain-deleted MED16 were transfected with the HIV-1 reporter. (E) The HIV-1 reporter was co-transfected with 1.5 μg of MED23, MED24, or Flag-tag control plasmids as additional controls. (F) Comparison of HIV-1 reporter luciferase activity between sh-NC and sh-MED16 293T cell lines. Firefly luciferase activity was normalized to Renilla luciferase activity. The luciferase expression levels between groups were compared using two-tail unpaired Student's t-test. The data are presented as the means ± SDs from n = 3 independent experiment. * $p < 0.05$, ** p -value < 0.01 and "ns": no significant.

panobinostat, a pan-HDAC inhibitor. Western blot analysis showed increased acetylation of histone H3 (H3ac) upon panobinostat treatment (**Fig S2A**). However, this treatment did not reverse MED16-mediated inhibition of HIV-driven luciferase activity (**Fig S2A**). These results suggest that MED16 does not rely on HDAC-mediated histone deacetylation to inhibit HIV-1 transcription and involves alternative mechanisms to repress HIV-1 transcription.

We then questioned whether the cis regulatory elements in the HIV-1 promoter are necessary for MED16-mediated inhibition. To address this question, we first examined whether the transactivating response region (TAR) plays a role in MED16-mediated HIV-1 inhibition. TAR is transcribed into TAR RNA, which recruits the host pTEFb-containing super elongation complex (SEC) to mediate the activation of viral protein Tat. UBP1 has been reported to restrict HIV-1 transcription at the level of elongation. A TAR RNA mutation HIV-1 reporter was created by replacing TAR loop nucleotides 31-34 UGGG with CAAA, which reduces binding affinity of Tat:P-TEFb and abolishes the function of TAR RNA. We observed that MED16 inhibited both wild-type and TAR-mutant reporter's transcription, while mutated TAR attenuated UBP1-mediated inhibition. This result suggested that MED16-mediated inhibition of HIV-1 is independent of the TAR-mediated HIV-1 transcriptional elongation.

HIV-1 proximal promoter has a string of GC-box that is bound by SP1, and an AP4 binding site between TATA-box and TSS. We then examined whether these two transcription factors interact with MED16 for transcription repression. A reporter driven by 4×GC-box was co-transfected with the MED16 expression plasmid, but no significant inhibition was observed compared with co-transfecting with the control vector, suggesting no cross-talk between MED16 and Sp1 (GC-box). Then we mutated the AP-4 binding motif CAGCTG to CAGTCG to abolish AP-4 binding. Overexpressing MED16 inhibited the luciferase activity, regardless AP-4 binding status, suggesting that MED16 and AP-4 don't work together to inhibit the HIV-1 transcription.

We then tested if the UBP1-MED16 inhibition of HIV-1 transcription is due to the direct UBP1 binding at the transcription started site (TSS). To test this hypothesis, two chimera reporters were created. Both are driven by a chicken β -ACTIN promoter. We replaced the native TSS site of the first reporter with the HIV-1 TSS contains the UBP1 binding site, while the second featured a chicken β -ACTIN promoter and a standard UBP1 binding site placed upstream of it. Various truncated mutants of MED16 and UBP1 were co-transfected with these reporters, respectively. In the case of the first chimera reporter, transcription was inhibited by full-length MED16 and MED16-bound UBP1 variants, but not by UBP1-binding deficient MED16 or MED16-binding deficient UBP1. Conversely, transcription of the second chimera reporter was not inhibited by any MED16 or UBP1 variants. Additionally, comparison of transcription activities between the two reporters revealed that the first chimera exhibited approximately 50% lower transcription activity than the second. These findings suggest that the positioning of the UBP1 binding site could be critical for MED16-mediated transcriptional repression.

We then investigated whether the position-dependent function of UBP1 also influenced UBP1-regulated endogenous gene expression across the human genome. To explore this, we analyzed the correlation between the relative binding position of UBP1 to TSS and gene transcription levels. We selected the top 50 upregulated and top 50 downregulated transcripts under UBP1 overexpression and analyzed the enrichment of the UBP1 motif around their promoters. Notably, UBP1-activated transcripts showed significant enrichment of the UBP1 motif upstream and downstream of the TSS, while UBP1-repressed transcripts exhibited motif enrichment specifically overlapping the TSS. Two UBP1-inhibited gene: *KIFC1* and *GTF2H4*, were found to have their TSS overlapping with UBP1 binding motif. Together, our data demonstrate that UBP1 may act as either a transcriptional activator or repressor, depending on distance of its binding site to TSS.

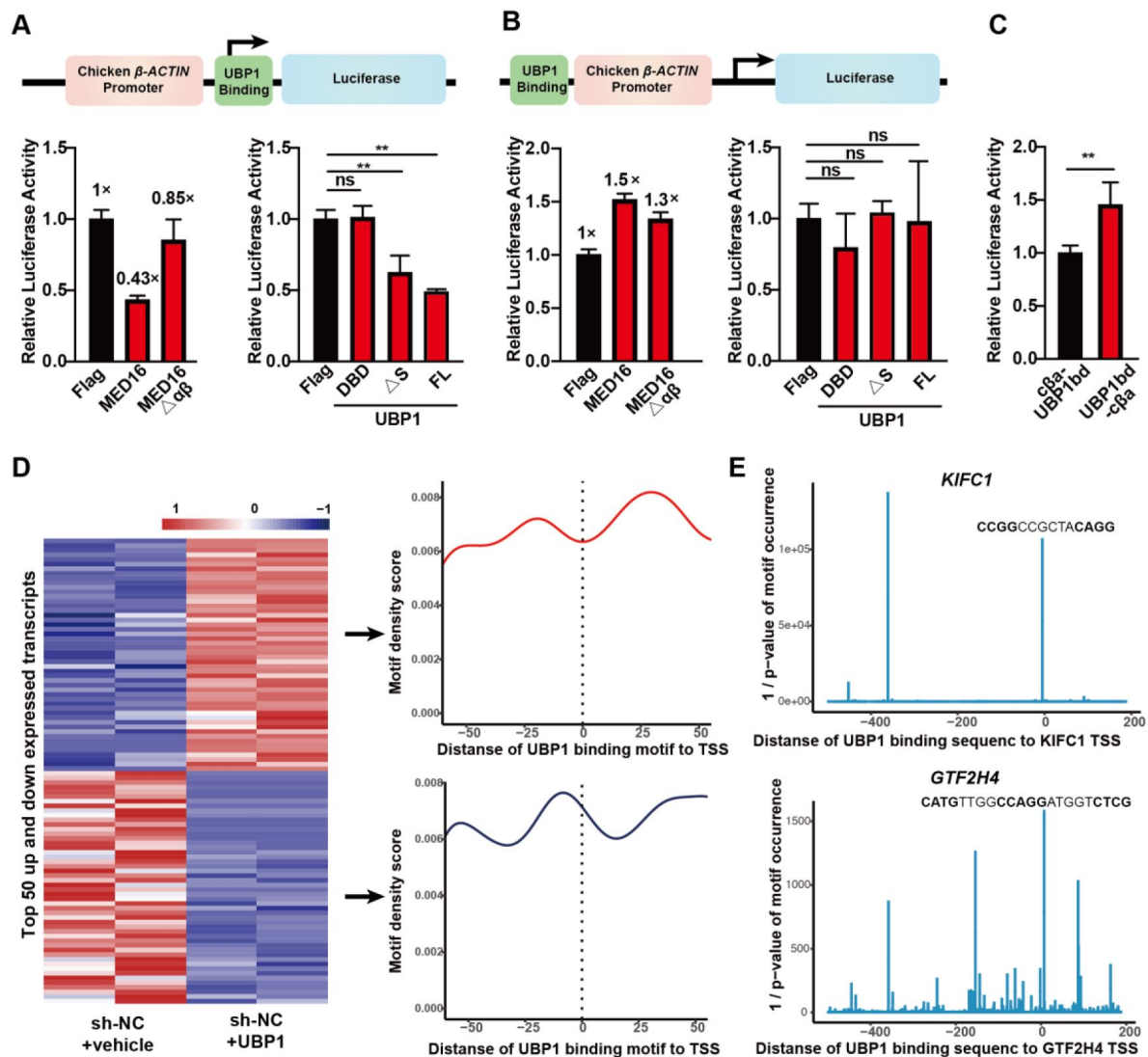


Figure 6.

UB1 binding site determines the UB1-MED16 inhibition in HIV-1 transcription

Schematic showing a chimera reporter (A) driven by the Chicken β -ACTIN promoter, followed by an HIV-1 UB1 binding site that also serves as the HIV-1 transcription start site and another chimera reporter (B) driven by the Chicken β -ACTIN promoter, a UB1 binding site modify from HIV-1 TSS placed upstream of it. 1 μ g of different truncation of MED16 or UB1 plasmid was co-transfected with the reporter. The Luciferase activity comparison between two chimera reporters were shown in (C). “c β a-UB1 bd” indicate the reporter that panel A described, and “UB1 bd-c β a” indicate the reporter that panel B described. Firefly luciferase activity was normalized to Renilla luciferase activity. The luciferase expression levels between groups were compared using two-tail unpaired Student’s t-test, The data are presented as the means $\pm$ SDs from $n = 3$ independent experiment. * $p < 0.05$, ** p -value < 0.01 and “ns”: no significant. UB1 motif enrichment analysis of top 50 upregulated or top 50 downregulated transcripts induced by UB1 were shown in (D). The transcripts changes levels are shown on the left as a heatmap, and average profile plots of UB1 motif density are shown on the right. UB1 motif occurrence analysis of UB1 downregulated gene *KIFC1* and *GTF2H4* were shown in (E). Motif occurrence were showed within the promoter region of the genes (-500 bp to 200 bp relative to the TSS).

MED16-UBP1 complex prohibited PIC formation of HIV-1 transcription

We wondered how UB1-MED16 binding within the proximal promoter might influence the HIV-1 transcription. To investigate this, we employed an *in vitro* immobilized template system. We first created nuclear extracts from three different cell lines derived from HT cells: specifically, wild type (WT), *Med16* knockout (Med16 KO), and overexpression (Med16 OE). HIV-1 core promoter (-78 to +60) templates were labeled with a single biotin molecule at their 5' ends and immobilized using streptavidin beads. These immobilized templates were then incubated with the NEs to form the pre-initiation complex (PIC). Transcription was initiated by adding four NTPs to stimulate transcription pause release and elongation. The binding proteins on the templates at each step were collected for western blotting, and the transcribed RNA was quantified (**Fig 7A**).

Med16 overexpression altered both PIC formation and transcription initiation stages of the immobilized template transcription process (**Fig 7B**). During the PIC forming stage, there was a decreased binding of CDK8, a key component of the Mediator kinase module. After transcription initiated, the signal for PolII pSer5, a well-known marker of transcription initiation, was reduced in MED16 OE NE compared to the WT (**Fig 7B**). Additionally, other Mediator subunits, such as MED12 from the kinase module, Med1 from the middle module, and MED23 from the tail module, exhibited decreased binding after transcription was initiated (**Fig 7B**). Moreover, Med16 overexpression prolonged the binding of general transcription factor TFIIB, which should be released after transcription initiation (61). Consequently, the Luciferase mRNA level in the MED16 OE was only 55% of that in the WT (**Fig 7C**). These results indicate that MED16 overexpression disrupts critical stages of HIV-1 transcription initiation, leading to a reduction in RNA production.

MED16 Maintains the HIV-1 latency

Because the MED16-UBP1 complex inhibits HIV-1 transcription (**Figure 5**), we explored whether it might also play a role in maintaining HIV-1 latency in CD4⁺ T cells. For this, we used the J-Lat 10.6 cell line, a Jurkat-based cell line infected with a pseudotyped HIV-1 strain (HIV/R7/E-/GFP) (Jordan et al., 2003). This cell line harbors the integrated HIV-1 copy in the second intron of the *SEC16* gene (Chung et al., 2020) and the integrated HIV-1 copy remains silenced under normal conditions. HIV-1 latency reversal can be monitored by GFP expression that is controlled by viral promoter and serves as a marker for viral activation (**Fig 8A**). To induce viral expression, we treated cells with the BET bromodomain inhibitor JQ1 that is known to reverse HIV-1 latency by antagonizing BRD4's suppression of the Tat-P-TEFb/SEC (Super Elongation Complex) interaction, (Li et al., 2013; Zhu et al., 2012).

We generated J-Lat 10.6 cell lines with retroviral MED16 expression and a control line infected with the MSCV (Murine Stem Cell Virus) retrovirus (**Fig 8B**). Prior to JQ1 treatment, both cell lines showed minimal GFP expression, confirming that MED16 manipulation alone did not alter HIV-1 latency (**Fig 8C**). After treatment with 5μM JQ1 for 48h, the HIV-LTR-driven GFP expression is observed in both cell lines, but is reduced in the MED16 OE cell line, with a 40% decrease in the percentage of GFP-positive cells percentage and 0.54-fold reduction in MFI (mean fluorescence intensity) compared to the control cell line (**Fig 8C**, **8D**, and **8E**). These findings suggest that MED16 not only inhibits HIV-1 transcription but also plays a crucial role in maintaining HIV-1 latency in CD4⁺ T cells.

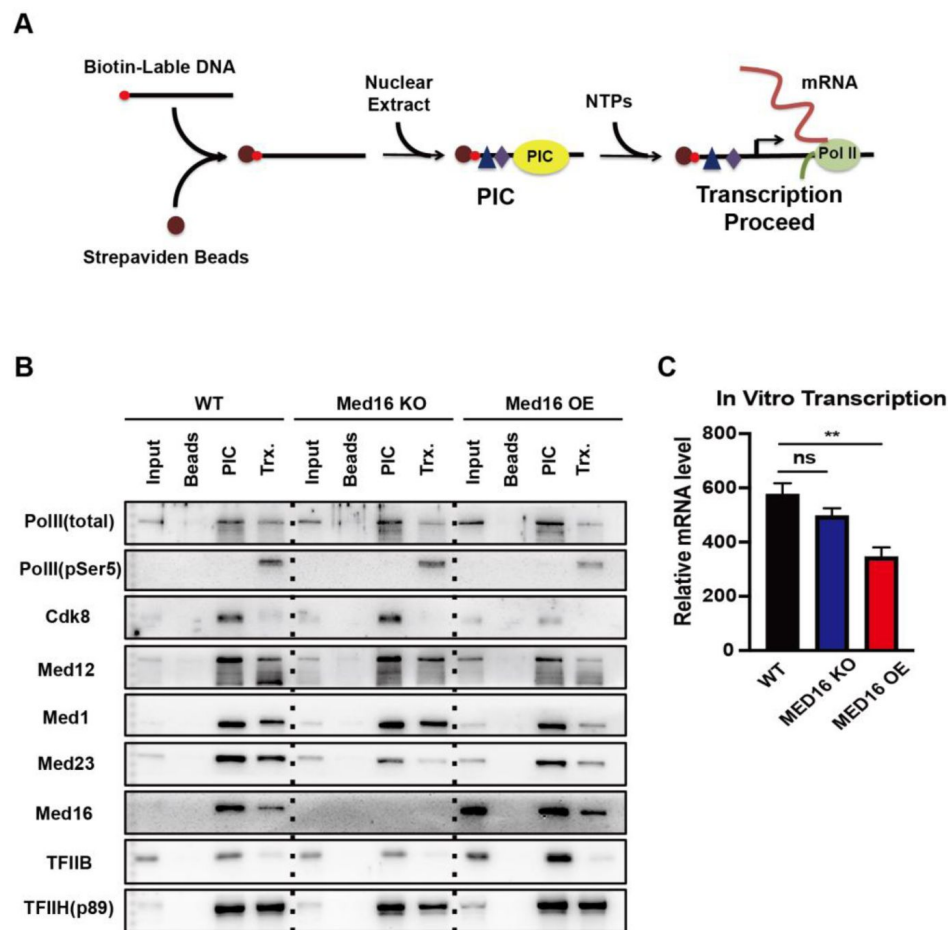


Figure 7.

MED16-UBP1 complex prohibited PIC formation of HIV-1 transcription

(A) Schematic of biotin-labeled immobilized template assay. (B) Immobilized template assay in wild-type, Med16 knockout and Med16 overexpression HT cell line. The binding proteins in individual steps were collected for immunoblotting. Beads: the streptavidin beads incubated with NE as a negative control; PIC: the pre-initiation step of transcription; Trx.: transcription initiation from the PIC step by adding NTPs. (C) qPCR quantification of generated RNA from different NE. The mRNA expression levels were compared between groups using two-tail unpaired Student's t-test. The values are presented as the means $\pm$ SDs from $n = 3$ independent experiment. ** p -value < 0.01 and "ns": no significant.

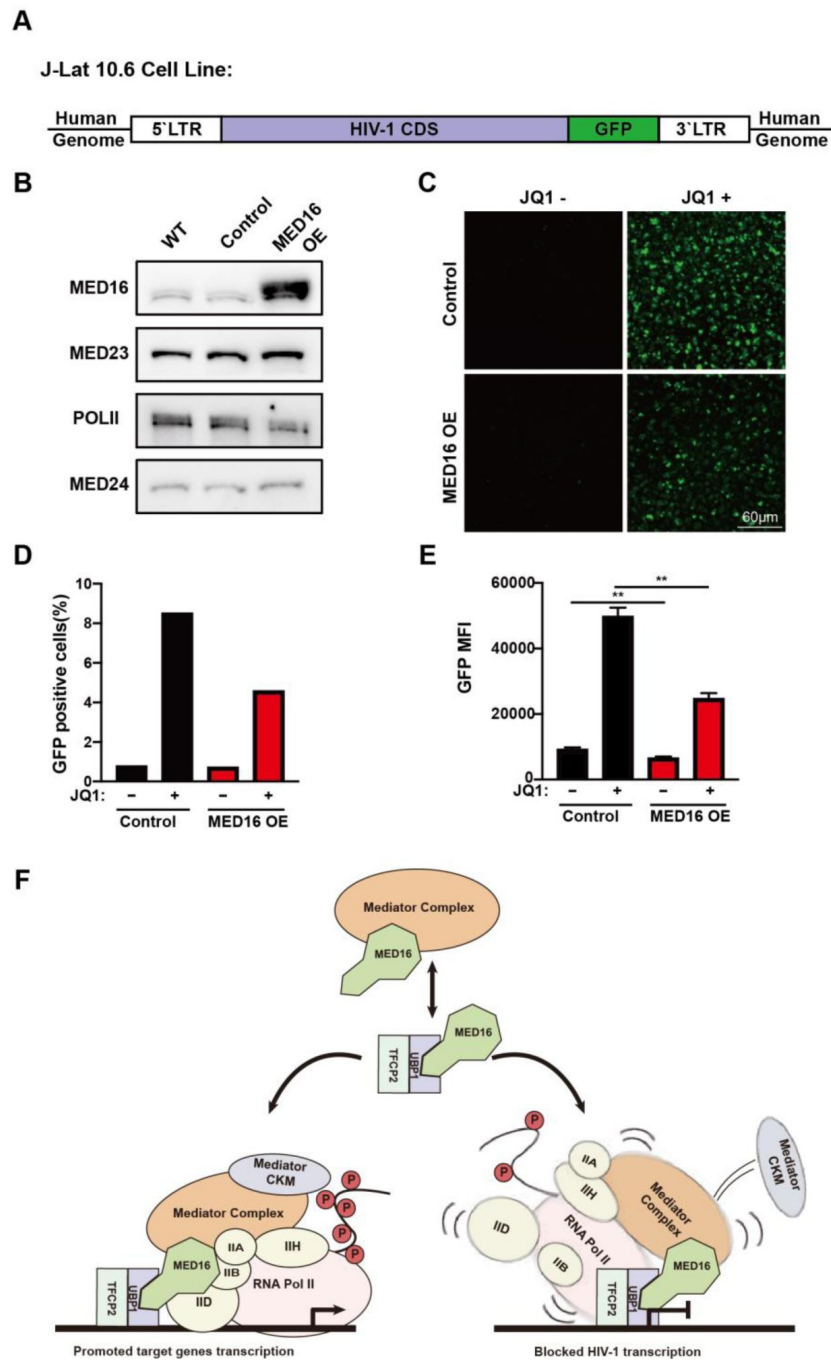


Figure 8.

MED16:UBP1 complex prohibited PIC formation of HIV-1 transcription

(A and B) Stable overexpression of MED16 (MED16 OE) in the J-Lat 10.6 cell line using MSCV retrovirus. Control: Control J-Lat 10.6 cell line generated via infection with a vehicle retrovirus. (C) Fluorescence microscopy images of MED16 OE and Control J-Lat 10.6 cells treated with or without 5 μ M JQ1 for 48 hours. Percentage of GFP-positive cells measured by FACS (D), and mean fluorescence intensity (MFI) quantified using the BioTek Synergy 2 system (E). The values are presented as the means $\pm$ SDs from $n = 3$ independent experiment. ** p -value < 0.01

Discussion

In this study we identified two transcription factors UBP1 and TFCP2 that interact with Mediator subunit MED16. This interaction activates an array of endogenous gene expression and inhibits HIV-1 transcription.

Utilizing gel-filtration chromatography and immunoprecipitation-mass spectrometry (IP-MS), we found a significant amount of MED16 co-elutes at the fraction between 443 kDa and 150 kDa where other core Mediator subunits are absent. Several transcription factors, including TFCP2, UBP1, and YY1 were co-immunoprecipitated with MED16 within these fractions (**Fig 1D**, **1E** and **1F**). Although these proteins were isolated from the Mediator complex in gel-filtration chromatography, they functionally relied on the Mediator complex since the WDR-deleted MED16, which cannot interact with UBP1, failed to repress HIV-1 activation (**Fig S3B**). MED16 interacts with UBP1 to activate the expression of an array of endogenous genes. In the previous studies, UBP1 deficiency results in lethality at midgestation due to the prominent angiogenic defect observed in the mutant placentas and allantoic blood vessels fail to penetrate deeply and branch into the complex embryonic vasculature characteristic of the normal placenta (**29**, **53**), which is consistent with our GO analysis that UBP1 and MED16 control angiogenesis gene expression. Unexpectedly, pulmonary surface molecule genes were also regulated by UBP1 and MED16. These genes encode pulmonary surfactant associated proteins, typically expressed in alveolar epithelial cells but silenced in HeLa cells. However, UBP1 overexpression activated them, likely due to the well-matched UBP1 motifs in their promoters.

Previous studies have demonstrated UBP1's role in inhibiting HIV-1 transcription through various mechanisms, including epigenetic and elongational regulation. However, our investigation proposes a novel mechanism where the MED16-UBP1 complex inhibits HIV-1 transcription by preventing PIC formation at the HIV-1 TSS. This inhibition is specific to the position of the UBP1 binding site, with transcription being repressed when UBP1 binds to its site within a TSS and activated when UBP1 binds to its site either upstream or downstream of a TSS. This finding highlights the importance of binding site positioning in transcriptional control, which is often overlooked in past studies. The position-dependent function of transcription factors (TFs) relative to the TSS is supported by a recent study demonstrating how TF binding site's spatial configuration can lead to distinct gene regulatory outcomes. (**62**) showed that the effect of TF binding on transcription initiation is highly position-dependent. TFs such as NRF1, NFY, and Sp1 activate or repress transcription depending on their exact position relative to the TSS. Our findings with the MED16-UBP1 complex align with this because UBP1 activates transcription when binding upstream or downstream of the TSS but represses it when binding directly at the TSS.

Targeted inhibition of HIV-1 transcription, particularly through enhancing the silencing activity of UBP1-MED16 at the viral transcription start site (TSS), presents a promising strategy for HIV-1 therapy. Many current approaches focus on inhibiting key regulatory proteins crucial for HIV-1 transcription (Ophinni et al., 2018; Peng et al., 2023; Yeh et al., 2020). By strengthening the UBP1-MED16 interaction or increasing MED16's affinity for the HIV-1 TSS, it may be possible to achieve more effective and stable transcriptional silencing, which could reduce viral expression over prolonged periods. In the future, small molecule drug screening can be conducted targeting the enhancement of this interaction, and the obtained small molecules could be used in combination with existing antiretroviral therapies (ART) to enhance therapeutic efficacy (Pai et al., 2021; Sorokina et al., 2023).

Interestingly, a small peptide fragment of MED16 has the opposing effect of activating HIV-1 transcription while potentially blocking the endogenous UBP1-MED16 interaction (**Fig S3C**). By manipulating the expressing form of MED16, we can create a multi-faceted therapeutic tool that

enables flexible regulation of HIV-1 transcription based on specific therapeutic needs. For instance, expressing the full-length MED16 could enhance the efficacy of antiretroviral therapies (ART), while expressing the MED16 $\alpha\beta$ -domain peptide could support "shock-and-kill" strategies (63) by reactivating latent HIV-1, thereby enabling immune clearance or targeted antiretroviral intervention in previously hidden reservoirs. Together, these dual capabilities of MED16 position it as a versatile approach in the treatment of HIV-1.

Method

Plasmids

The human *MED16*, *YY1*, *TFCP2*, *TFCP2L1* cDNAs were obtained from BRICS (Bio-Research Innovation center SUZHOU). The human *UBP1* and mouse *Med16* cDNAs were cloned from HeLa cells and HT cells respectively. Then full-length cDNA and truncations were cloned into 3×Flag-CMV-10 (Sigma-Aldrich-Aldrich) and pET-28b plasmids. The HIV-1 core promoter sequence were synthesized directly and clone into pGL3-Basic plasmid. All mutations on plasmid were generated by using FASR Site-Directed Mutagenesis Kit (Tiangen; Beijing; China). siMED16 and Ctrl shRNA oligonucleotides were cloned into pSiren-RetroQ (Clontech).

Cell culture

HeLa and 293T cells were maintained in Dulbeccòs modified Eaglès medium (DMEM) containing 10% (v/v) FBS and 1% P/S. The HT cell line used in this study was an immortalized pancreatic cancer cell line derived from a *KRAS*^{G12D} *P53*^{+/-} mouse and was generous provided by professor Xiaofei Yu in Fudan University. J-Lat 10.6 cell line was generous provided by professor Huanzhang Zhu in Fudan University. HT and J-Lat 10.6 cell were maintained in RPMI-1640 medium containing 10% (v/v) FBS and 1% P/S. All cells were maintained in the incubator with 5% CO₂ at 37°C.

Western blot and real-time PCR assays

Method for Western blot and real-time PCR assay have been described previously (64). The primers for real-time PCR are listed in Table S1. Antibody for Western blot include the following MED23 (Abcam, ab200351), MED1 (Bethyl Lab, A300-793A), MED24 (Bethyl, A301-472A), MED6 (Santa Cruz, sc-9434), MED12 (Bethyl Lab, A300-774A), MED16 (Bethyl Lab, A303-668A), CDK8 (Abcam, ab115155), UBPI (Proteintech, 67318-1-Ig), TFCP2 (Proteintech, 15203-1-AP), YY1 (Proteintech, 66281-1-Ig), P300 (Abcam, ab14984), HDAC1 (Santa Cruz, sc-7872), PolII (Abcam, ab816), PolII-pSer5 (Millipore, 04-1572), TFIIB (Santa Cruz, sc-274D), TFIIF (Santa Cruz, sc-293), H3 (Abcam, ab1791), H3ac (Abcam, ab4729), β -Actin (Proteintech, 66009-1-Ig), GAPDH (Proteintech, 60004-1-Ig), Flag-tag (Sigma-Aldrich, F1804), His-tag (Proteintech, 66005-1-Ig), GST-tag (Invitrogen, 10004D).

Nuclear extract Preparation and gel-filtration chromatography assay

Preparation of un-dialyzed HeLa cell nuclear extract was described previously (65). Gel-filtration molecular weight markers (Sigma-Aldrich, MWGF1000) were loaded onto a Superose 6 10/300GL column to determine the void volume (Blue dextran ~2000 kDa) and elution volume of protein standards (Thyoglobulin 669 kDa, Apoferritin 443kDa, Alcohol Dehydrogenase 150 kDa, Albumin 66 kDa). HeLa nuclear extract was applied to the Superose 6 10/300GL column and then was run in Buffer D (20 mM HEPES (pH 7.9), 300 mM KCl, 10% glycerol, 0.2 mM EDTA, 10 mM β -mercaptoethanol, and 1 mM PMSF). Fractions of 500 μ l were collected from the column and precipitated using trichloroacetic acid (TCA). Every third fraction was used for Western blot assay.

Co-immunoprecipitation (Co-IP) assay

For transient co-transfection, 293T cells plated in 10 cm dishes (90% confluency) were transfected with 10 µg of each plasmid using Lipofectamine 2000 (Invitrogen) according to the manufacturer's protocol. After 48 h, the cells were harvested and lysed in lysis buffer (20 mM HEPES pH7.5, 150mM NaCl, 1mM EDTA, 2.5mM EGTA, 0.3% Tween-20) with protease inhibitors. Lysates were subjected to immunoprecipitation with anti-Flag M2 beads (Sigma-Aldrich, A2220) and incubated at 4°C overnight, following by washing in lysis buffer three times. Then 50 µl 1× SDS loading buffer was added to the beads and boiled at 99°C for 10 min for Western blot analysis with the indicated antibodies.

For endogenous Co-IP, 293T cells were seeded in 10 cm dishes and cultured under appropriate conditions. At nearly 80% confluence, the cells were harvested and lysed in lysis buffer with protease inhibitors. Lysate was centrifuged at 4°C, 13200 rpm for 10 minutes, and the supernatant was incubated with 4µg antibodies to MED16 (Bethyl, A303-668A), CDK8 (CST, 17396s), MED1 (Bethyl, A300-793A), MED12 (Bethyl, A300-774A) overnight. Then, 20 µl of Dynabeads Protein G (Invitrogen) was added and incubated for 2 hours at 4°C. The beads were then washed with lysis buffer (with 0%, 5%, 10% 1,6-hexanediol) three times and boiled with SDS loading buffer for Western blot assay with indicated antibodies.

As for the Co-IP experiment in HeLa nuclear extracts (NE) or gel-filtration fractions, the HeLa NE was first diluted with an equal volume of D300 buffer containing 20 mM HEPES (pH 7.9), 300 mM KCl, 10% glycerol, 0.2 mM EDTA, 10 mM β-mercaptoethanol, and 1 mM PMSF. Next, 4 µg of antibody was added directly to the diluted HeLa NE or gel-filtration fractions, followed by incubation at 4°C overnight. After the overnight incubation, 20 µL of Dynabeads Protein G (Invitrogen) was added and incubated for an additional 2 hours at 4°C. The beads were then washed three times with D300 buffer and boiled in SDS loading buffer for Western blot analysis with the specified antibodies.

IP-MS and analysis

After immunoprecipitation, the protein samples were separated by SDS-PAGE, running at 80 V for about 15 minutes. The electrophoresis was halted once the bromophenol blue dye front had traveled 0.5 cm into the resolving gel. The gel was then stained with Coomassie Brilliant Blue R250 to visualize protein bands, which were subsequently excised and diced into small fragments (1 mm³). These gel fragments underwent sequential washes: twice with a 50% acetonitrile (ACN) solution in 50 mM ammonium bicarbonate (NH₄HCO₃), followed by three washes with 10 mM NH₄HCO₃ and one wash with pure ACN to remove salts and detergents.

After dehydration with 100% ACN, the gel pieces were treated with 10 mM dithiothreitol (DTT) for 1 hour at 37 °C to reduce disulfide bonds. The DTT solution was then removed, and the gel fragments were dehydrated again with ACN. Next, 50 mM iodoacetamide (IAM) was added to alkylate free thiol groups, with incubation for 30 minutes in the dark. The gel pieces were washed three times with 10 mM NH₄HCO₃ and once with ACN before final dehydration.

For enzymatic digestion, the gel fragments were incubated overnight at 37 °C with trypsin (3 ng/µl in 10 mM NH₄HCO₃). The resulting peptides were extracted using a gradient ACN elution and analyzed by liquid chromatography-tandem mass spectrometry (LC-MS/MS) on an Orbitrap Fusion™ Lumos™ system.

Generation of Med16 KO cell line using CRISPR-Cas9

To generate *Med16* knockout cell lines, the pX330-mCherry plasmid, containing the CRISPR-Cas9 machinery and target-specific guide RNAs, was used. A guide RNA sequence as reported in a previous study(66) was cloned into the pX330-mCherry vector. Cells were transfected with the pX330-mCherry plasmid using Lipofectamine 2000 (Invitrogen) according to the manufacturer's protocol. After 48 hours, single mCherry-positive cells were sorted by fluorescence-activated cell sorting (FACS) into individual well of 96-wells plate. The sorted cells were then cultured and expanded. Individual clones were screened for gene knockout by sequencing and Western blotting to confirm successful editing.

Retrovirus infection

Stable cell lines with retroviral-mediated knockdown or overexpression were generated following the manufacturer's protocol (Clontech) as described previously (13 [DOI](#)). Specific siRNA target sequences were designed using Thermo Scientific's siRNA selection tools. Retroviruses were produced by co-transfecting 293T cells with pSiren-RetroQ (for knockdown) or pMSCV-puro (for overexpression) vectors along with the pCL10A1 helper plasmid, using Lipofectamine 2000 (Invitrogen). After 48 hours, the culture medium containing retroviruses was collected, supplemented with 20 µg/ml polybrene (Sigma-Aldrich), and filtered through a 0.45 µm filter. The filtered retroviruses were then added to target cells, which were subjected to spin infection at 2,500 rpm for 1.5 hours at 30°C. Twenty-four hours after infection, cells were selected with puromycin in 293T at 1 µg/ml and HeLa cells at 2.5 µg/ml.

Luciferase reporter assay

293T cells or HeLa cells were plated in a well of 12-wells plate (90% confluency) and transfected with 150 ng of luciferase reporter plasmid along with 50 ng pRL-TK (Promega) using Lipofectamine 2000 (Invitrogen). At 36-48h post transfection, the cells were harvested and subjected to dual-luciferase reporter assays according to the manufacturer's protocol (Promega)

RNA-seq and data analysis

Total RNA was extracted from cells lysed with TRIZOL. We used a commercial RNA library kit to prepare the RNA-seq library according to the manufacturer's specifications (VAHTS® Universal V8 RNA-seq LibraryPrep Kit) from Illumina. Briefly, mRNA was captured with mRNA capture beads before fragmentation. The RNA fragments were reverse transcribed and subjected to library amplification for Illumina (Novaseq 6000) sequencing.

For RNA-seq data analysis, the filtered clean reads were mapped to the human reference genome hg38 with HISAT2. Read counts for each gene were then quantified using HTSeq-count. Differential expression analysis was performed in R using DESeq2, which normalized the read counts and identified significantly differentially expressed genes.

Motif analysis

Motif enrichment analysis was performed using the closest known homolog of UBP1: the Tcfcp2l1 (GSE11431) position weight matrix (PWM) from the HOMER motif database (v4.9.1). Sequences and the Tcfcp2l1 PWM were submitted to the FIMO algorithm (MEME Suite v5.5.8) to scan for enriched motifs. This analysis generated base pair-resolution motif enrichment scores and associated p-values across all queried sequences.

In vitro transcription assays with immobilized template

The immobilized template assay was carried out based on the previous study(67). Biotin-labeled PCR products were generated from the HIV-1 reporter plasmid with biotin attached to the upstream end. These templates contain the HIV-1 core promoter region (-78 to +60) followed by the luciferase CDS. 6 µg of each biotinylated template was bound to 300 µg of Dynabeads M-280 Streptavidin (Dyna) according to the manufacturer's instructions using a DynalMag™-2 (Dyna) for assistance. The immobilized templates were then resuspended in 100 µL of blocking buffer (20 mM HEPES, pH 7.9, 0.1 M KCl, 10% glycerol, 6 mM MgCl₂, 2.5 mM DTT, 50 mg/mL BSA). After a 15-minute incubation at room temperature, 100 µL of nuclear extract from either WT, MED16-overexpressing, or *Med16* KO HT cells was added and incubated at 30°C for 30 minutes to allow formation of pre-initiation complex (PIC). The immobilized templates were washed three times with buffer D containing 0.1% Triton X-100. The proteins bound to the templates were eluted with SDS gel loading buffer and analyzed by 10% SDS-PAGE.

For in vitro transcription, an NTP mix (25 mM each) was added to stimulate transcription after PIC formation, and the reaction was incubated at 30°C for 1 hour. The supernatant, containing the transcribed RNA products, was collected for qPCR quantification. The immobilized templates were again washed three times with buffer D plus 0.1% Triton X-100, and the proteins bound to the templates were eluted with SDS gel loading buffer and subjected to 10% SDS-PAGE.

Quantification and statistical analysis

For qPCR and luciferase assays, the means and SEMs were calculated from at least three independent experiments. Statistical significance was assessed using GraphPad Prism software, with data presented in graphs as means ± SDs. Student's t-test was used for comparisons between two groups, with significance levels defined as * $p < 0.05$ and ** $p < 0.01$.

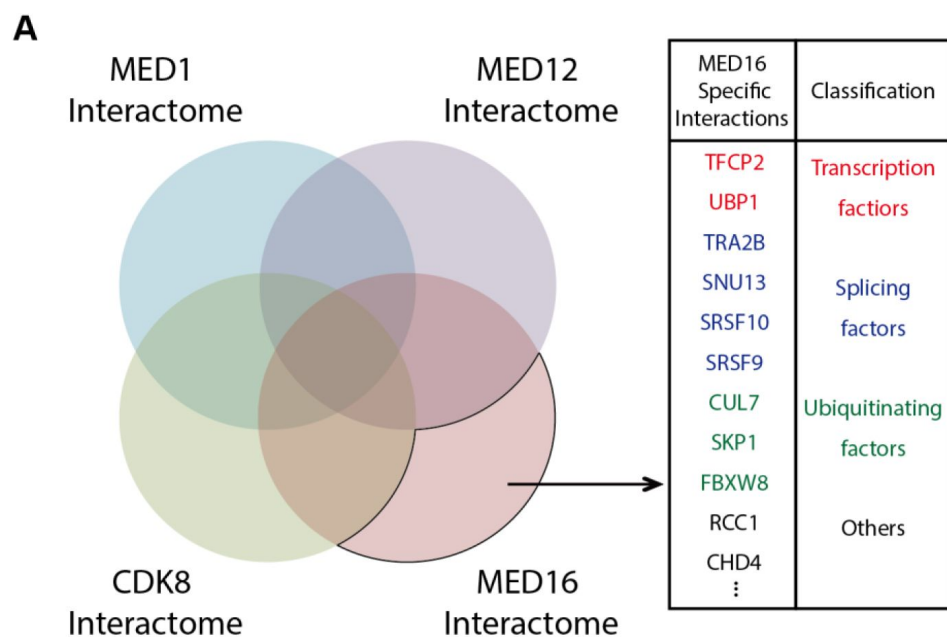


Figure S1.

Venn diagram of MED1, MED12, CDK8 and MED16 interacting proteins identified by IP-MS.

MED16 specific interacting proteins were highlight on the right.

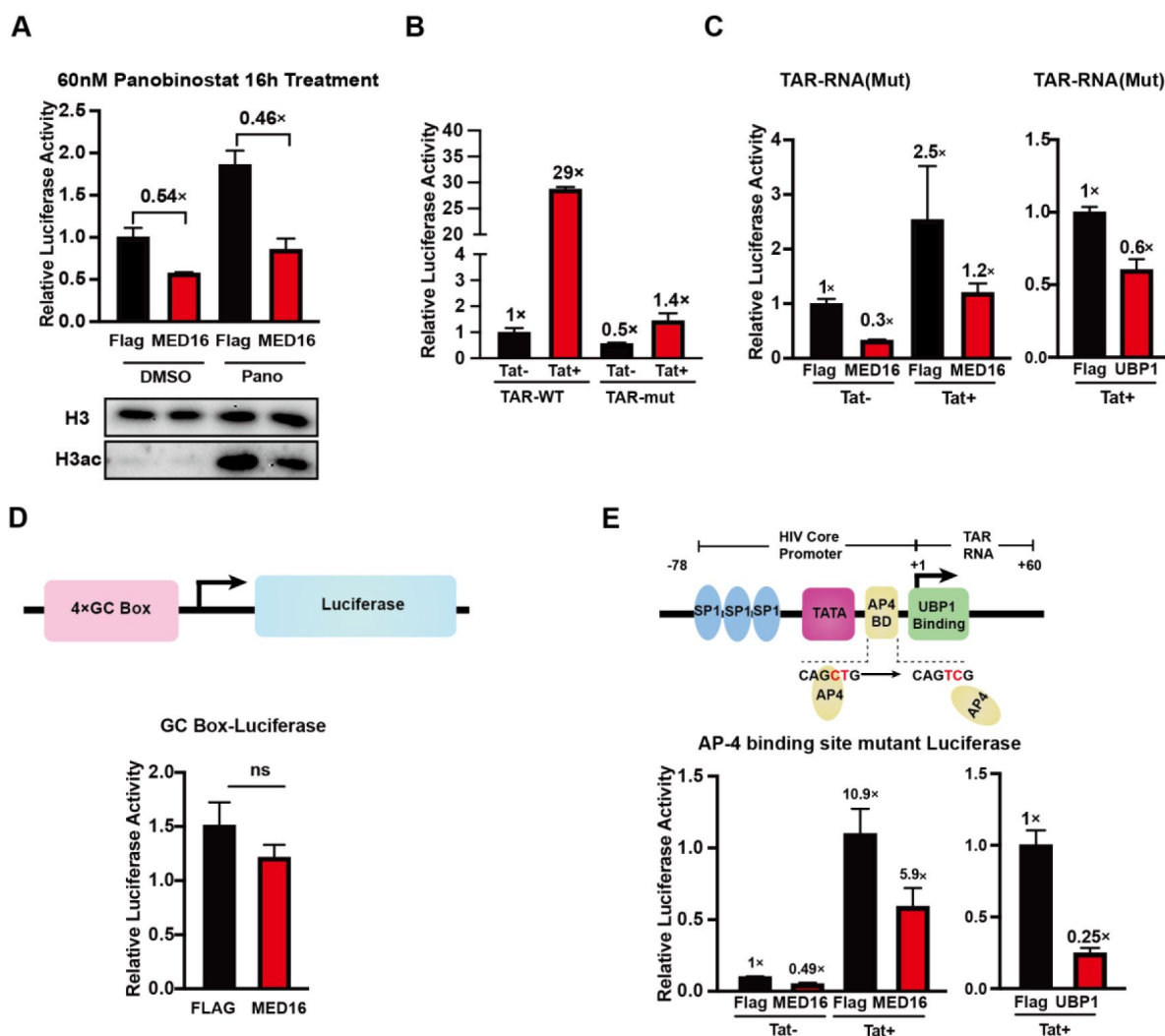


Figure S2.

Identifying the MED16-controlled regulatory elements at the HIV-1 promoter

293T cells were transfected with 300 ng of either Flag control, MED16 overexpression, or UBPI overexpression plasmid, along with the indicated reporter plasmids. (A) Cells were co-transfected with an HIV-1 reporter plasmid, then treated with or without 60 nM Panobinostat for 16 hours. Deacetylation inhibition was validated by western blot analysis (shown below). (B) Co-transfection with a TAR RNA mutant HIV-1 reporter and 50 ng of Tat overexpression plasmid, to assess the mutation's effects. (C) Co-transfection with a TAR RNA mutant HIV-1 reporter to evaluate regulation by Flag control, MED16, or UBPI overexpression. (D) Co-transfection with a 4xGC box reporter plasmid to examine MED16's impact on GC box activity. (E) Co-transfection with an AP-4 binding site mutant HIV-1 reporter to assess MED16's effect on the AP-4 site. The data are presented as the means $\pm$ SDs from $n = 3$ independent experiments; "ns": no significant.

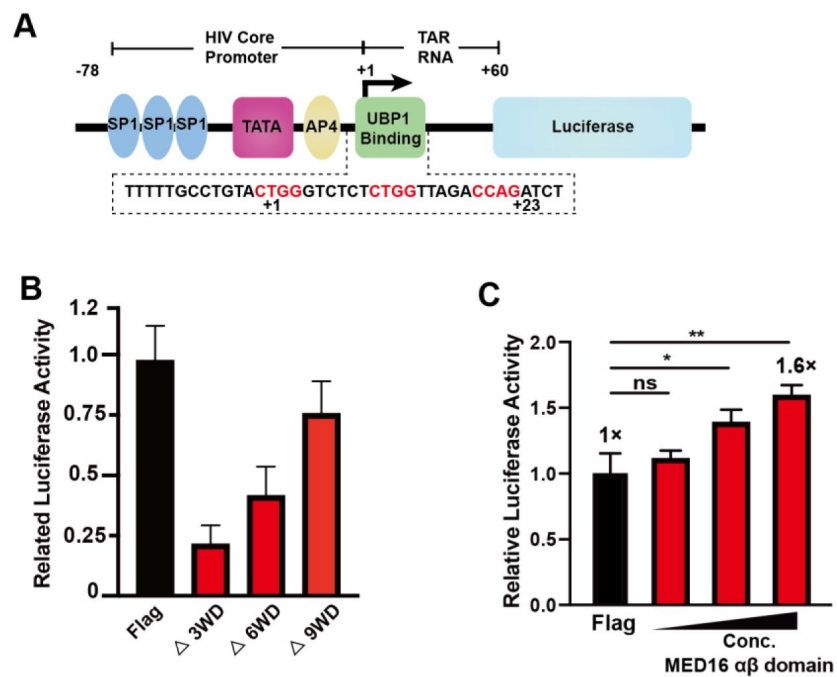


Figure S3.

Effect of MED16 truncations co-transfected with HIV-1 reporter

(A) Diagram of the HIV-1 promoter, with UB1 binding consensus sequences highlighted in red. (B) 293T cells were transfected with 1 μ g of MED16 truncation plasmids (Δ 3WD, Δ 6WD, and Δ 9WD, corresponding to constructs 2, 3, and 4 in [Fig. 2A](#)) along with the HIV-1 reporter plasmid shown in (A). (C) 293T cells were transfected with increasing amounts (0.3 μ g, 0.5 μ g, and 1.0 μ g) of MED16 $\alpha\beta$ -domain peptide overexpression plasmid together with the HIV-1 reporter. The data are presented as the means $\pm$ SDs from $n = 3$ independent experiment. * p -value <0.05 , ** p -value <0.01 and "ns": no significant.

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Reviewer #1 (Public Review):

Summary:

Characterization of a dissociable Mediator subunit implicated in cellular pathways, particularly lung alveolar function, and HIV latency is conceptually interesting.

Strengths:

The strengths of this study are:

(1) Demonstration of MED16 dissociation from the core Mediator complex and formation of a subcomplex containing MED16, upstream-binding protein 1 (UBP1), and transcription factor cellular promoter 2 (TFCP2) by elegant biochemical fractionation and immunoblotting analysis.

(2) Defining nine N-terminal WD-40 repeats (WDRs) of MED16 as a Mediator-incorporating module and the C-terminal $\alpha\beta$ -domain (157 amino acids) important for interaction with the UBP1-TFCP2 heterodimeric complex.

(3) Illustration of a weak hydrophobic interaction between MED16 and the Mediator core that could be disrupted by 1,6-hexanediol, but not by its 2,5-hexanediol isomer nor by high salt (500 mM NaCl) disruption.

(4) Classification of UBP1-upregulated cellular genes typically containing binding sites flanking the transcription start site (TSS) in contrast to UBP1-downregulated genes often containing a TSS-overlapping UBP1-binding site

(5) Presenting evidence for Mediator complex-dissociated free MED16-repressed HIV promoter activity through functional association with UBP1 and showing bromodomain-containing protein 4 (BRD4) inhibitor JQ1 that potentially disrupts BRD4-inhibited HIV-1 transcription elongation could lead to reversal of HIV-1 latency.

Weaknesses:

Nevertheless, foreseeable weaknesses include:

(1) No clear demonstration of MED16-UBP1-TFCP2 indeed forming a trimeric core subcomplex in regulating cellular gene transcription and HIV-1 promoter inhibition

(2) No validation of transcriptomic datasets and pathways identified.

(3) Use of mostly artificial reporter gene constructs and non-HIV host cells (e.g., human 293T embryonic kidney cells, human HeLa cervical cancer cells, and mouse HT pancreatic cancer cells) for examining MED16/UBP1-regulated HIV transcription.

(4) Inconsistent use of 293T and HeLa cells in the characterization of dissociated MED16 interaction with UBP1 and TFCP2.

(5) In vitro transcription using immobilized DNA templates was not performed to a high standard, thus failing to convincingly show MED16/UBP1-inhibited HIV-1 transcription preinitiation complex formation.

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Reviewer #2 (Public Review):

Summary:

The article from Zheng et al. proposes an interesting hypothesis that the Med16 subunit of Mediator detaches from the complex, associates with transcription factor UBP1, and this complex activates or represses specific sets of genes in human cells. Despite my excitement upon reading the abstract, I was concerned by the lack of rigor in the experimental design. The only statement in the abstract that has some experimental support is the finding that Med16 dissociates from the Mediator and forms a subcomplex, but the data shown remain incomplete.

Strengths:

The authors have preliminary evidence that a stable Med16 complex may exist and that it may regulate specific sets of genes.

Weaknesses:

The experiments are poorly designed and can only infer possible roles for Med16 or UBP1 at this point. Furthermore, the data are often of poor quality and lack replication and quantitation. In other cases, key data such as MS results aren't even shown. Instead, we are given a curated list of only about 6 proteins (Figure S1), a subset of which the authors chose to pursue with follow-up experiments. This is not the expected level of scientific process.

(1) The data supporting the Med16 dissociation and co-association with UBP1 are incomplete and not convincing at this stage. According to the Methods and text, the gel filtration column was run with "un-dialyzed HeLa cell nuclear extract" and eluted in 300mM KCl buffer. The extracts were generated with the Dignam/Roeder method according to the text. Undialyzed, that means the extract would be between 0.4 - 0.5M NaCl. Under these high salt conditions (not physiological), it's possible and even plausible that Mediator subunits could separate over time. This caveat is not mentioned or controlled for by the authors. Because a putative Med16 subcomplex is a foundational point of the article, this is concerning.

The data are incomplete because a potential Med16 complex is not defined biochemically. The current state suggests a smaller Med16-containing complex that may also contain UBP1 and other factors, but its composition is not determined. This is important because if you're going to conclude a new and biologically relevant Med16 complex, which is a point of the article, then readers will expect you to do that.

Equally concerning are the IP-western results shown in Figure 1. In my opinion, these experiments do nothing to support the claims of the authors. The authors use hexanediols at 5% or 10% in an effort to disrupt the Mediator complex. Assuming this was weight/volume, that means ~400 to 800mM hexanediol solution, which is fairly high and can be expected to disrupt protein complexes, but the effects haven't been carefully assessed as far as I'm aware. The 2,5 HD (Figure 1B) experiments appear to simply contain greater protein loading, and this may contribute to the apparent differential results. In fact, in looking at the data, it seems that all MED subunits probed show the same trend as Med16. They are all reduced in the 1,6HD experiment relative to the 2,5 HD experiment. But it's hard to know, because replicates weren't completed and quantitation was not done. There aren't even loading controls. Other concerns about the IP-Western experiments are outlined in point 2.

(2) At no point do the authors apply rigorous methods to test their hypothesis. Instead, methods are applied that have been largely discredited over time and can only serve as preliminary data for pilot studies, and cannot be used to draw definitive conclusions about protein function.

a) IP-westerns are fraught with caveats, especially the way they were performed here, in which the beads were washed at relatively low salt and then eluted by boiling the beads in loading buffer. This will "elute" bound proteins, but also proteins that non-specifically interact with or precipitate on the beads. And because Westerns are so sensitive, it is easy to generate positive results. It's just not a rigorous experiment.

b) Many conclusions relied on transient transfection experiments, which are problematic because they require long timeframes, during which secondary/indirect effects from expression/overexpression will result. This is especially true if the proteins being artificially expressed/overexpressed are major transcription regulators, which is the case here. It is simply impossible to separate direct from indirect effects with these types of experiments. Another concern is that there was no effort to assess whether the induced protein levels were near physiological levels. Protein overexpression, especially if the protein is a known regulator of pol2 transcription (e.g., UBP1 or Med16), will create many unintended consequences.

c) Many conclusions were made based upon shRNA knockdown experiments, which are problematic because they require long timeframes (see above point), which makes it nearly

impossible to identify effects that are direct vs. indirect/secondary/tertiary effects. Also, shRNA experiments will have off-target effects, which have been widely reported for well over a decade. An advantage of shRNA knockdowns is that they prevent genetic adaptation (a caveat with KO cell lines). A minimal test would be to show phenotypic rescue of the knockdown by expressing a knockdown-resistant Med16 (for example), but these types of experiments were not done.

d) Many experiments used reporter assays, which involved artificial, non-native promoters. Reporters are good for pilot studies, but they aren't a rigorous test of direct regulatory roles for Med16 or other proteins. Reporters don't even measure transcription directly. In fact, no experiment in this study directly measures transcription. An RNA-seq experiment was done with overexpressed or Med16 knockdown cells, but these required long timeframes and RNA-seq measures steady-state mRNA, which doesn't test the potential direct effects of these proteins on nascent transcription.

e) The MS experiments show promise, but the data were not shown, so it's hard to judge. The reader cannot compare/contrast the experiments, and we have no indication of the statistical confidence of the proteins identified. How many biological replicate MS experiments were performed?

(3) The data are over-interpreted, and alternative (and more plausible) hypotheses are ignored. Many examples of this, some of which are alluded to in the points above. For example, Med16 loss or overexpression will cause compensatory responses in cells. An expected result is that Mediator composition will be disrupted, since Med16 directly interacts with several other subunits. Also in yeast, the Robert, Gross, and Morse labs showed that loss of Med16/Sin4 causes loss of other tail module subunits, and this would be expected to cause major changes in the transcriptome. The authors also mention that yeast Med16/Sin4 "alters chromatin accessibility globally" and this would be expected to cause major changes in the transcriptome, leading to unintended consequences that will make data analysis and identification of direct Med16 effects impossible. The unintended consequences will be magnified with prolonged disruption of MED16 levels in cells (e.g., longer than 4h). These unintended consequences are hard to predict or define, and are likely to be widespread given the pivotal role of Mediator in gene expression. One unintended consequence appears to be loss of pol2 upon Med16 over-expression, as suggested by the western blot in Figure 8B. I point this out as just one example of the caveats/pitfalls associated with long-term knockdowns or over-expression.

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Reviewer #3 (Public Review):

Summary:

There are two major flaws that fundamentally undermine the value of the study. First, nearly all the central conclusions drawn here rely on the unfounded assumption that the effects observed are direct. No rigorous cause-and-effect relationships are established to support the claims. Second, the quality of the experimental data is substandard. Collectively, these concerns significantly limit any advances that might be gained in our understanding of the UBP1 pathway or Mediator function.

Weaknesses:

(1) The decrease in 1,6-hexanediol-treated cells of MED16 is modest, variable, not quantified, and internally inconsistent. For example, in Figure 1A, 1,6-hexanediol treatment should not have an impact on the level of the protein being directly IP. For MED12 (and CDK8 and MED1

to a lesser extent), 1,6-hexanediol treatment alters the level of the target protein in the IP. Along these lines, Figure 1A shows a no 1,6H-D dependent decrease in MED1 or MED12 levels in the CDK8 IP, whereas Figure 1B does show a decrease. Figure 1A shows no 1,6H-D dependent decrease in CDK8 levels in the MED1 IP, whereas Figure 1B shows a dramatic decrease. MED24 levels in the MED12 IP increase upon 1,6H-D in Figure 1A, but decrease in Figure 1B. Internal inconsistencies of this nature persist in the other Figures.

(2) Undermining the value of Figure 1E/F, UBP1 and TFCP2 may also associate with the small amount of MED16 in the 2MDa fractions. This is not tested, and therefore, the conclusion that they just associate with the dissociable form of MED16 is not supported.

(3) Domain mapping studies in Figure 2 are overinterpreted. Since the interactions could be indirect, it is not accurate to conclude "Therefore, the N-terminal WDR domain of MED16 is crucial for its integration into the Mediator complex, while the C-terminal $\alpha\beta$ -domain is essential for interacting with UBP1-TFCP2. "

(4) A close examination of Figure 2C undermines confidence in the association studies. The bait protein in lanes 5-8 should be equal. Also, there is significant binding of GST to UBP1 and TFCP2, in roughly the same patterns as they bind to GST-MED16 $\alpha\beta$. The absence of input samples makes the results even more difficult to interpret.

(5) The domain deletion mutants are utilized throughout the manuscript as evidence of the importance of the UBP1-MED16 interaction. However, in Figure 2F lanes 7 and 8, the delta-S mutant binds MED16 as well as full-length UBP1. This undermines much of the subsequent data and conclusions about specificity.

(6) Even if the delta-S mutant were defective for MED16 binding, the result in Figure 3B does not "confirm that MED16 is required for the transcriptional activity of UBP1, ". Removal of that domain may have other effects.

(7) As Mediator is critical for the activation of many genes, it is not accurate to assume that the impact of its deletion in Figure 3E/F demonstrates a direct requirement in UBP1-driven transcription. This could easily be an indirect effect.

(8) Without documenting the relative protein expression levels in Figure 3G/H, conclusions cannot be drawn about the titration experiments, nor the co-expression experiments. These findings are likely the result of squelching or some form of competition that is not directly related to the UBP1-mediated transcription. A great deal of validation would be required in order to support the model that these effects are a result of MED16 overexpression sequestering UBP1 away from holo-Mediator.

(9) The lack of any documentation of expression levels for the various ectopic proteins in the majority of Figures, renders mechanistic claims meaningless (Figures 3, 4, 5, 6, 7, S2, S3). This is particularly relevant since the model presented for many of the results invokes concentration-dependent competition.

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